

The Two Strategies — Equations & a Worked Monthly Walkthrough

autoPortfolio: technical reference

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0. Notation

Symbol	Meaning
r_i	daily (or monthly) return of asset i
w_i	portfolio weight on asset i
$\hat{\sigma}_t$	estimated (trailing, realized) volatility at time t , annualized
e_t	exposure at t — fraction of the book that is invested (rest is cash); $e_t \in [0, 1]$
g	gross exposure of the currency book ($= 1$)
c	per-name cost (half-spread) in basis points (bps); 1 bp = 0.01%

Two layers everywhere: **weights** (how the invested money is split) and **exposure** (how much is invested at all). They are computed separately.

1. Equity strategy — the equations

1.1 Selection (quarterly)

Score every eligible point-in-time member by liquidity (dollar volume), then take the top $N = 30$ subject to a **sector cap** (at most $M = 5$ per sector). Formally the book is

$$\mathcal{B} = \underset{\text{sector cap } M}{\text{top-N}} \text{ (liquidity score)}.$$

1.2 Weights — equal weight

Each selected name gets the same weight:

$$w_i = \frac{1}{N} = \frac{1}{30} \approx 3.33\%, \quad \sum_{i \in \mathcal{B}} w_i = 1.$$

1.3 Exposure — the 15% volatility target

This is the **equity exposure equation**. Measure the book’s trailing realized volatility and scale exposure so risk stays near a constant target $\sigma_{\text{tgt}} = 15\%$:

$$\boxed{e_t^{\text{eq}} = \min\left(1, \frac{\sigma_{\text{tgt}}}{\hat{\sigma}_{t-1}}\right)} \quad \hat{\sigma}_{t-1} = \text{std}(r_{t-21}, \dots, r_{t-1}) \times \sqrt{252}.$$

- $\hat{\sigma}$ uses only **past** returns (a 21-trading-day window), lagged one step, so there is no look-ahead.
- The $\min(1, \cdot)$ means we **never lever** — exposure only ever falls below 100%; the shortfall $1 - e_t$ sits in cash.
- Calm markets ($\hat{\sigma} < 15\%$) $\Rightarrow e_t = 1$ (fully invested). Turbulent markets ($\hat{\sigma} > 15\%$) $\Rightarrow e_t < 1$ (de-risked).

The final invested weight on name i is $e_t \cdot w_i$.

1.4 The “water-filling” cap algorithm

Where it is used: enforcing a **per-name cap** c_{max} when a weighting scheme would otherwise concentrate too much in one name (e.g. the sentiment-tilt experiments). The deployed **equal-weight** book never triggers it, because $1/30 = 3.3\%$ is already below any sensible cap — but here is the exact rule, because plain “clip then renormalize” both **violates** the cap and **flattens** distinct weights.

Given raw weights w (summing to 1) and cap c_{max} , repeat until no weight exceeds the cap:

Let $O = \{i : w_i > c_{\text{max}}\}$. If $O = \emptyset$, stop.

Excess: $E = \sum_{i \in O} (w_i - c_{\text{max}})$.

Cap them: $w_i \leftarrow c_{\text{max}} \quad \forall i \in O$.

Redistribute proportionally to the uncapped names:

$$w_j \leftarrow w_j + E \cdot \frac{w_j}{\sum_{k \notin O} w_k} \quad \forall j \notin O.$$

Each pass moves the “overflow” above the cap line down and pours it into the names still below the line (hence *water-filling*), **preserving their relative ordering**. It converges in a few passes and always yields $\sum_i w_i = 1$ with every $w_i \leq c_{\max}$ (feasible whenever $N \cdot c_{\max} \geq 1$).

2. Currency strategy — the equations

2.1 The signal — carry

For each currency, carry is its 3-month interest rate minus the US dollar's, annualized in percent:

$$\text{carry}_i = r_i^{3M} - r_{\text{USD}}^{3M}.$$

Positive $\Rightarrow$ higher-yielding than the dollar (a *long* candidate); negative $\Rightarrow$ lower-yielding (a *short* candidate).

2.2 Weights — rank, then long/short, dollar-neutral

Rank all currencies by carry. Go **long the top n** and **short the bottom n** , equal weight within each sleeve:

$$w_i = \begin{cases} +\frac{g/2}{n} & i \in \text{top-}n \text{ (highest carry)} \\ -\frac{g/2}{n} & i \in \text{bottom-}n \text{ (lowest carry)} \\ 0 & \text{otherwise} \end{cases}$$

with gross $g = 1$. Then

$$\sum_i w_i = 0 \quad (\text{dollar-neutral}), \quad \sum_i |w_i| = g = 1.$$

With $n = 3$: each long weight $= +\frac{0.5}{3} = +0.1667$, each short $= -0.1667$.

2.3 Return accounting — carry-inclusive

Holding a foreign currency (funded in dollars) earns the **spot move plus the interest differential** over the holding period $\Delta t = \frac{1}{12}$ (monthly):

$$R_i = \underbrace{\frac{S_{i,t}}{S_{i,t-1}} - 1}_{\text{spot return}} + \underbrace{\frac{\text{carry}_i}{100} \cdot \Delta t}_{\text{interest earned}}$$

The book return, net of trading cost, is

$$R_t^{\text{book}} = \sum_i w_i R_i, \quad \text{cost}_t = \sum_i |w_i - w_i^{\text{prev}}| \cdot \frac{c_i}{10^4},$$

where c_i is that currency's half-spread in bps (majors ≈ 5 , EM ≈ 30).

2.4 Exposure — the HMM crash overlay (optional)

This is the **currency exposure equation**. A 2-state (calm / stress) Gaussian Hidden Markov Model is fit on crash features (cross-currency realized vol; safe-haven vs high-beta return spread). Let $P(\text{stress} \mid \text{obs} \leq t)$ be its **filtered** probability (forward algorithm, frozen parameters — uses only past data). Then

$$e_t^{\text{fx}} = \text{clip}(1 - P(\text{stress} \mid \text{obs} \leq t - 1), 0, 1)$$

so exposure falls toward 0 as the stress probability rises. The final book return with the overlay is

$$R_t^{\text{net}} = e_t^{\text{fx}} \cdot R_t^{\text{book}} - |e_t^{\text{fx}} - e_{t-1}^{\text{fx}}| \cdot \frac{c}{10^4}.$$

3. Worked example — a full monthly reallocation

Using the **current** data (latest month).

3.1 Currency book — step by step

Step 1 — signal. From FRED, compute each currency’s carry vs USD. The extremes of the ranking:

Currency	carry vs USD	Role
ZAR (S. Africa)	+3.34%	long
MXN (Mexico)	+2.99%	long
HUF (Hungary)	+2.21%	long
SEK (Sweden)	−1.82%	short
JPY (Japan)	−2.45%	short
CHF (Switzerland)	−3.81%	short

Step 2 — weights. $n = 3$, so each long = $+\frac{0.5}{3} = +0.1667$, each short = -0.1667 (sum = 0, gross = 1).

Step 3 — realize the month. For each name, $R_i = \text{spot ret} + \text{carry}/1200$:

Currency	w_i	spot ret	+ carry accrual	$= R_i$	$w_i R_i$
CHF	−0.1667	−0.24%	−0.318%	−0.558%	+0.093%
JPY	−0.1667	+1.55%	−0.204%	+1.346%	−0.224%
SEK	−0.1667	+0.66%	−0.151%	+0.509%	−0.085%
HUF	+0.1667	+0.13%	+0.185%	+0.315%	+0.053%
MXN	+0.1667	+1.34%	+0.249%	+1.589%	+0.265%
ZAR	+0.1667	+2.31%	+0.278%	+2.588%	+0.431%

Step 4 — book return. Sum the last column:

$$R^{\text{book}} = 0.093 - 0.224 - 0.085 + 0.053 + 0.265 + 0.431 = +\mathbf{0.53\%} \text{ (this month).}$$

What this shows: the short-JPY leg **lost** (−0.224%) because the yen rallied (a safe-haven move) — that is exactly the carry risk. But the EM longs (MXN, ZAR) more than covered it. In a true risk-off month, the yen would spike **and** the EM longs would crash together — that is the negative-skew “carry crash,” and it is why the exposure overlay (e_t^{fx}) exists.

Step 5 — costs & exposure. Turnover is tiny (the ranking rarely changes), so cost is a fraction of a bp. If the HMM reads “calm,” $e^{\text{fx}} = 1$ and the net return is $\approx +0.53\%$; if it read stress at $P = 0.4$, exposure would be $e = 1 - 0.4 = 0.6$ and we would realize $0.6 \times 0.53\% = 0.32\%$.

3.2 Equity book — step by step

Step 1 — holdings (quarterly). Select the 30 most-liquid point-in-time names, capped at 5 per sector, equal weight $w_i = 1/30 = 3.33\%$ each.

Step 2 — exposure (checked monthly/weekly). Measure trailing 21-day realized vol. Currently

$$\hat{\sigma}_{t-1} = 16.9\% \Rightarrow e_t^{\text{eq}} = \min\left(1, \frac{0.15}{0.169}\right) = \mathbf{0.89}.$$

So the book is **89% invested, 11% cash**. Each name's live weight is $0.89 \times 3.33\% = 2.97\%$.

Step 3 — if markets turn. Say vol jumps to 25% next month:

$$e^{\text{eq}} = \min\left(1, \frac{0.15}{0.25}\right) = 0.60 \Rightarrow \text{cut to 60\% invested, 40\% cash,}$$

each name to $0.60 \times 3.33\% = 2.0\%$. The trades to get there are the only turnover; the holdings themselves don't change until the next quarterly rebalance.

4. One-glance summary of the equations

Equity weight — equal weight on the sector-capped selection:

$$w_i = 1/N.$$

Equity exposure — 15% volatility target:

$$e_t^{\text{eq}} = \min(1, \sigma_{\text{tgt}}/\hat{\sigma}_{t-1}), \quad \sigma_{\text{tgt}} = 0.15.$$

Water-fill cap — cap the over-limit names at c_{max} , pour the excess E into the uncapped names in proportion to their weight ($w_j \propto w_j$), repeat until all $w_i \leq c_{\text{max}}$.

Currency weight — long/short the top/bottom- n by carry, dollar-neutral:

$$w_i = \pm \frac{g/2}{n} \text{ (else 0),} \quad \sum_i w_i = 0, \quad \sum_i |w_i| = g.$$

Currency return — carry-inclusive:

$$R_i = \left(\frac{S_{i,t}}{S_{i,t-1}} - 1\right) + \frac{\text{carry}_i}{1200}, \quad R^{\text{book}} = \sum_i w_i R_i - \text{cost}.$$

Currency exposure — HMM crash overlay:

$$e_t^{\text{fx}} = \text{clip}(1 - P(\text{stress}), 0, 1).$$